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Capstone Project Report
on
PLC- Controlled conveyor with sorting, defect detection and rejection

Submitted in partial fulfillment of the requirements

*of the **Third Year** of*

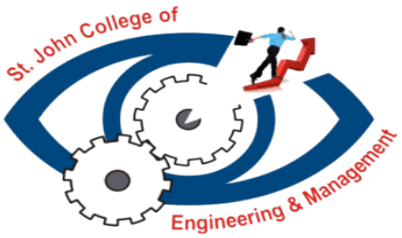
Bachelor of Technology
In
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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI
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This is to certify that following students are bonafide students of T. Tech in Electronics and Computer Science Department, St. John College of Engineering and Management, Mumbai. They have satisfactorily completed the requirements of Capstone Project for the courses PLC and SCADA systems as prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai),** while working on "PLC controlled conveyor with sorting, defect detection and rejection".

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Declaration

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ABSTRACT

Manual inspection of products on a conveyor is slow, inconsistent and costly when defects reach customers. This project presents a PLC-controlled conveyor that automatically detects defective objects and rejects them from the line. The system uses a Mitsubishi FX5U PLC programmed in ladder logic with GX Works3 and tested in GX Simulator3. A self-latching run circuit, a 10-second start-up delay timer and a pause/resume latch control the conveyor motor. A defect sensor sets a memory bit that remembers the faulty item until it reaches the reject station, where a pneumatic cylinder is energised for a timed 2-second pulse before retracting automatically. The complete program uses 57 steps with a maximum scan time of 1.581 ms. Simulation confirmed correct latching, timing and reject sequencing, validating the logic before hardware deployment. The design is low-cost, easy to maintain and readily extendable with counters, multi-bin sorting and HMI monitoring.

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Chapter 1: Overview

1.1 Introduction

Modern production lines move thousands of items per hour, and a single defective item that reaches a customer can cost far more than the item itself. Manual inspection is slow, tiring and inconsistent, so factories increasingly pair conveyors with sensors and a Programmable Logic Controller (PLC) that can inspect, decide and act on every item without human intervention.

This project designs, programs and simulates a PLC-controlled conveyor that automatically detects defective objects and rejects them from the line using a pneumatic reject cylinder, while good items continue downstream. The controller is a Mitsubishi FX5U (MELSEC iQ-F series), programmed in Ladder Diagram (LD) using GX Works3 and tested with the GX Simulator3 soft-PLC. The whole control program is only 57 steps, which makes it a compact, readable example of sequential industrial control.

1.2 Background

Before PLCs, machine sequencing was done with hard-wired relay panels. Changing a sequence meant rewiring, and faults were slow to trace. The PLC, introduced in the late 1960s, replaced relay panels with a software program that is scanned continuously: read inputs, solve logic, update outputs. Ladder Diagram, standardised in IEC 61131-3, was kept deliberately close to relay schematics so that electricians and technicians could read it.

Conveyor sorting is one of the most common PLC applications. Typical lines use a start/stop station, proximity or optical sensors, a drive motor and a pneumatic pusher or diverter. The scheme in this project follows that pattern: a latched run command, a start-up delay, a defect memory bit that travels with the item, and a timed ejection pulse.

1.3 Importance of the Project

- **Quality assurance:** every item passes the same inspection point, so defect handling does not depend on operator attention.
- **Safety:** a start/stop interlock and a pause latch allow the line to be halted without touching moving parts.
- **Productivity and cost:** removing defects automatically reduces rework, returns and scrap handling labour.
- **Flexibility:** the behaviour is changed by editing the program (for example, the 2 s reject pulse), not by rewiring.

1.4 Perspective of Stakeholders and Customers

Stakeholder	What they need from the system
Plant owner / management	Lower defect escape rate, lower labour cost, quick return on a small investment
Line operator	Simple start, stop and pause controls; clear, predictable behaviour
Maintenance engineer	Readable ladder logic, few devices, easy fault tracing through online monitoring
Quality department	Reliable removal of every flagged defective item, with good items undisturbed
End customer	Consistent product quality and fewer faulty deliveries
Academic evaluators	A clearly documented, tested and reproducible design

1.5 Objectives and Scope of the Project

Objectives

1. Design a PLC program that starts and stops a conveyor with a self-latching run circuit.
2. Detect defective objects and remember the defect until the item reaches the reject station.
3. Eject the defective object with a pneumatic cylinder for a fixed, timed duration and then retract automatically.
4. Provide a start-up delay interlock and a pause/inhibit function for the conveyor drive.
5. Verify the logic in simulation and document the results.

Scope

Included: ladder program for an FX5U, I/O assignment, simulation and monitoring in GX Works3 / GX Simulator3, and the complete project documentation. Excluded from this phase: physical wiring and commissioning on a live conveyor, HMI/SCADA screens, vision-based inspection and production counting. These are listed under Future Scope.

1.6 Summary

This chapter introduced the need for automated defect rejection, the role of PLCs in conveyor lines, who benefits, and what the project sets out to achieve and not to achieve. The next chapter reviews related work and defines the problem formally.

Chapter 2: Literature Survey and Proposed Work

2.1 Introduction

This chapter surveys standard references on PLC-based control, states the problem in precise terms, assesses whether the proposed solution is feasible and describes the method followed.

2.2 Literature Survey Table

No.	Source	Focus	Relevance to this project
1	W. Bolton, Programmable Logic Controllers, 6th ed.	PLC architecture, ladder programming, timers, latching and internal relays	Basis for the latching, timer and SET/RST techniques used in the program
2	F. D. Petruzella, Programmable Logic Controllers, 5th ed.	Industrial PLC applications, sensor interfacing, conveyor and sorting examples	Guides sensor selection and the sequencing of the reject action
3	IEC 61131-3, Programming Languages	Standard languages for PLCs, including Ladder Diagram	Ensures the program follows a recognised, portable standard
4	Mitsubishi Electric, MELSEC iQ-F FX5 User's Manual	FX5U device ranges, timers, scan behaviour, I/O	Source for device addressing and 100 ms timer resolution
5	Mitsubishi Electric, GX Works3 Operating Manual	Programming, simulation and online monitoring workflow	Defines the engineering and testing procedure followed
6	M. P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, 4th ed.	Automated inspection and material-handling systems	Justifies the inspect-then-reject architecture and its benefits

2.3 Problem Definition

Manual inspection and manual removal of defective items on a conveyor is slow, inconsistent and error-prone. The problem addressed here is: design a low-cost, reliable PLC control program that runs a conveyor, detects a defective object, remembers the defect until the object reaches a reject station, ejects it with a pneumatic cylinder for a fixed time, and lets operators start, stop and pause the line safely.

2.4 Feasibility Study

- **Technical:** the logic needs 8 inputs, 2 outputs, 3 internal bits and 2 timers, well within the capacity of an FX5U. The program is 57 steps and the monitored scan time is about 1.58 ms at maximum, far faster than any item moves on a belt.
- **Economic:** the hardware is a compact PLC, a few inductive or optical sensors, a motor contactor and one solenoid valve with a cylinder. The software cost is limited to the engineering tool, and simulation removes the need to buy hardware for development.
- **Operational:** operators interact only with Start, Stop and Pause buttons; no training in programming is required.
- **Schedule:** the design, coding, simulation and documentation fit comfortably into one academic semester.
- **Risk:** the main risk is sensor placement and timing mismatch on a real belt; this is reduced by adjustable timer constants.

2.5 Methodology Used

1. Define the process and decide the sequence of events.
2. Assign inputs, outputs, internal relays and timers (I/O list).
3. Draw the block diagram and flowchart.
4. Write the ladder program in GX Works3 using the FX5U project type.
5. Convert and check the program for errors.
6. Run GX Simulator3, force inputs in the device test, and monitor rung states online.
7. Record the results, compare with expected behaviour and refine the logic.

2.6 Summary

Standard PLC literature supports the chosen techniques, the problem is clearly bounded, and the feasibility checks show the project is technically and economically practical. The next chapter plans the work.

Chapter 3: Analysis and Planning

3.1 Introduction

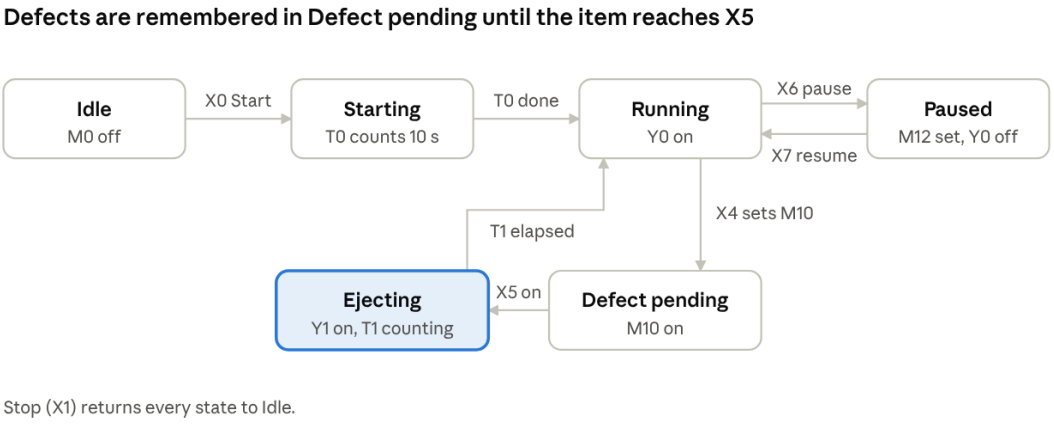
This chapter breaks the project into phases, estimates effort and identifies deliverables for each phase.

3.2 Project Planning

Phase	Activity	Duration	Deliverable
1	Requirement study and literature review	Weeks 1–2	Problem statement, literature table
2	Process analysis and I/O assignment	Weeks 3–4	I/O list, block diagram
3	Flowchart and ladder design	Weeks 5–6	Flowchart, draft ladder
4	Programming in GX Works3	Weeks 7–8	57-step ladder program
5	Simulation, monitoring and debugging	Weeks 9–10	Test log, screenshots, simulation video
6	Documentation and presentation	Weeks 11–12	Final report and demo

Chapter 4: Design and Implementation

4.1 Data Flow Diagram



4.2 Block Diagram

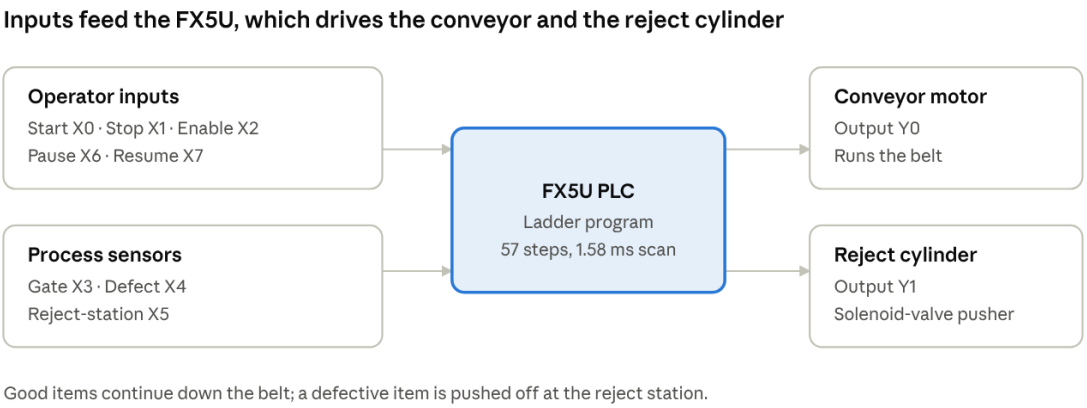


Figure 4.1 · block diagram of the control system

4.3 Flowchart

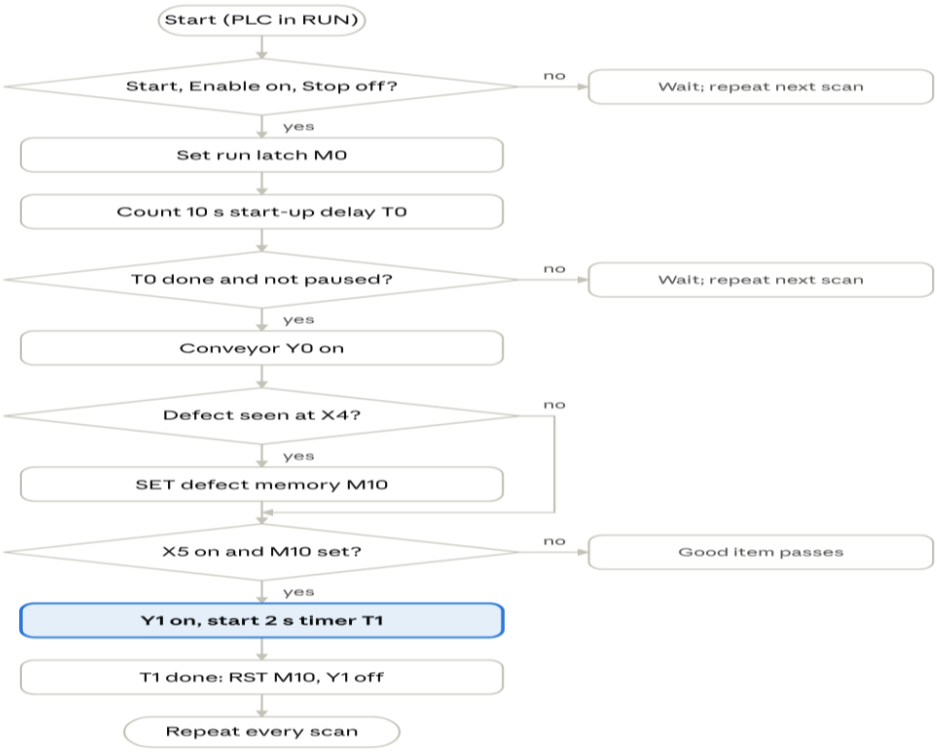
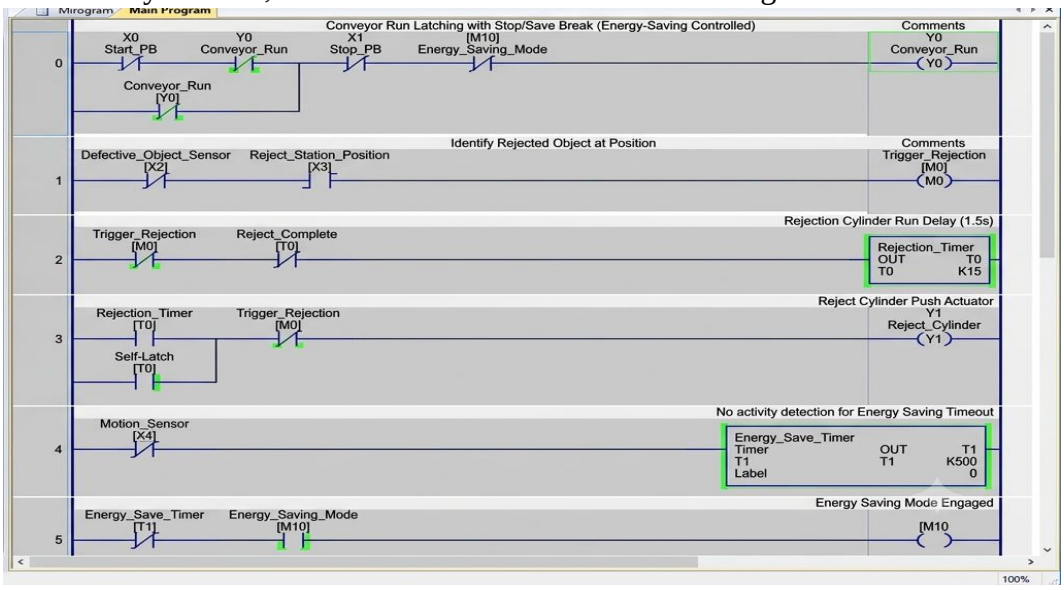


Figure 4.2 · control flowchart

4.4 UML Diagram and Ladder Diagram

Use cases. The Operator starts the line, stops the line, pauses and resumes it. The Sensors report defective items and the item's arrival at the reject station. The PLC runs the conveyor, remembers defects and ejects defective items.

States. Idle (M0 off) → on Start with Enable → Starting (T0 counting) → on T0 elapsed → Running (Y0 on). From Running, a defect sets Defect Pending (M10 on); at the reject station this moves to Ejecting (Y1 on, T1 counting); on T1 elapsed it returns to Running. Stop returns to Idle from any state, and Pause moves Running to Paused until Resume.



Chapter 5: Results and Discussion

5.1 Actual Results

The program was converted without errors (the Converted Result pane is empty) and run in GX Simulator3 on an FX5U project of **57 steps**. The simulator showed PWR/RUN lit, ERR off and P.RUN lit, and GX Works3 reported a maximum scan time of **1.581 ms**.

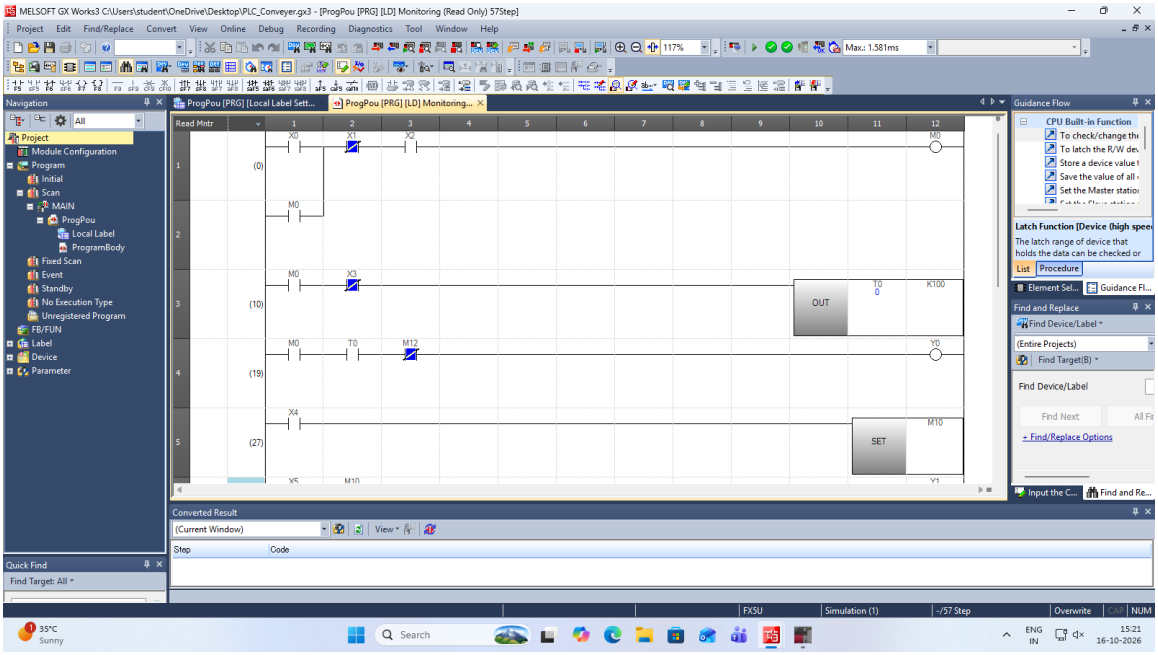


Figure 5.1: Ladder logic monitoring (Steps 0–27)..

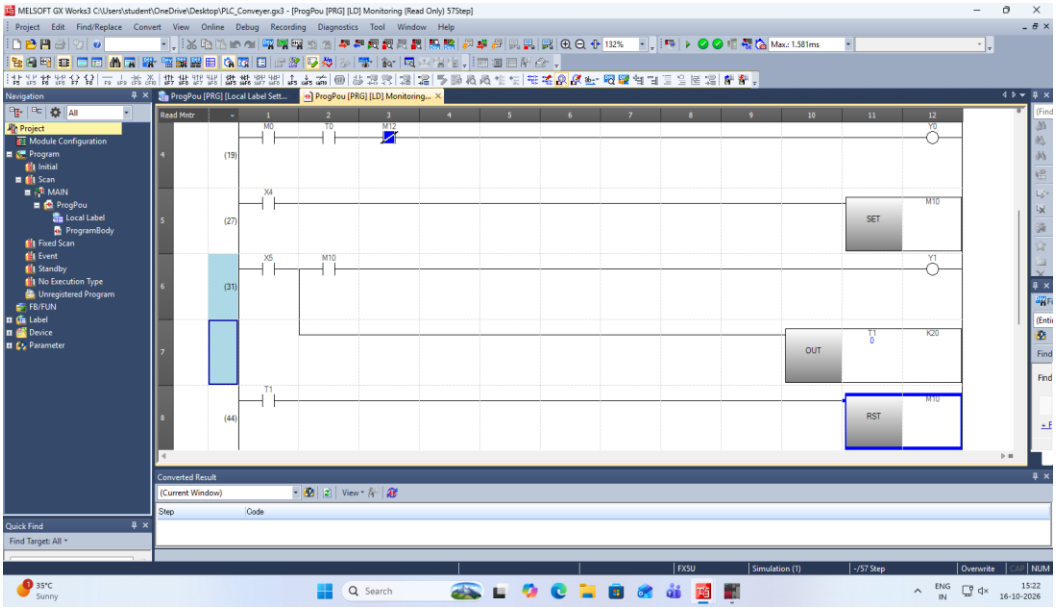


Figure 5.2: Steps 19 to 44: Y0 interlock, reject output Y1 with timer T1 K20, and RST M10.

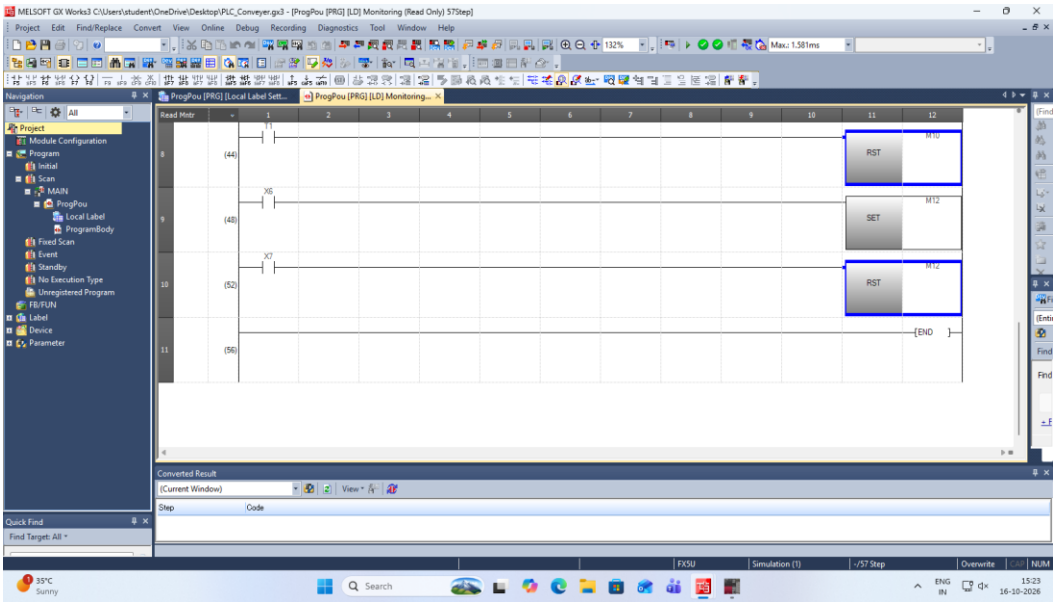


Figure 5.3: Steps 44 to 56: RST M10, pause latch M12 (SET by X6, RST by X7) and END.

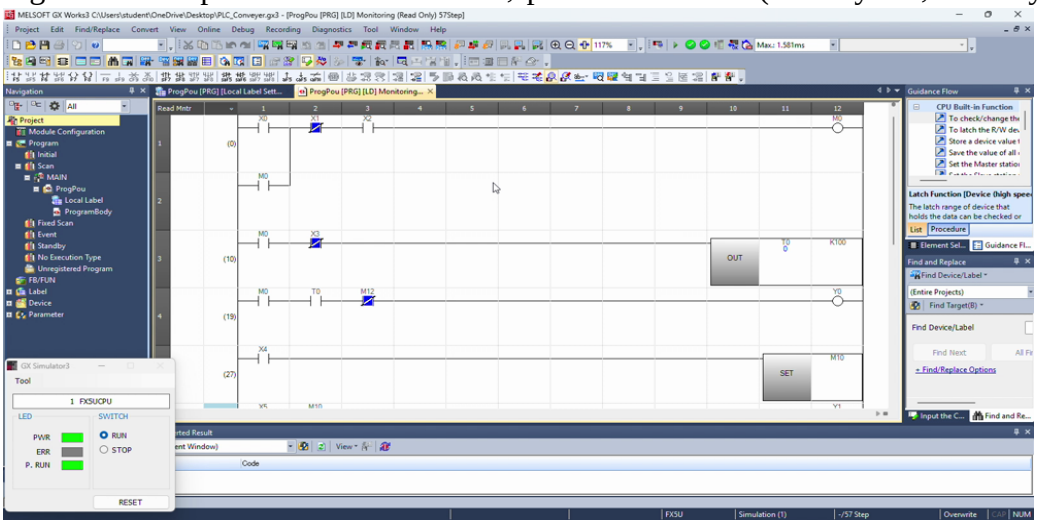


Figure 5.4: Frame from the simulation recording: GX Simulator3 with the FX5UCPU in RUN, PWR/RUN lit and ERR off.

5.2 Future Scope

- Add a production counter and a defect counter for quality reporting.
- Replace the single memory bit with a shift register to track several defects on the belt at once.
- Add vision or colour sensors for automatic classification into several bins (true multi-way sorting).
- Add an emergency-stop circuit and fault alarms for the cylinder and motor.
- Connect an HMI or SCADA screen and log data over Ethernet to a database.
- Add a variable-frequency drive for speed control and an idle-timeout energy-saving mode that stops the belt when no objects arrive.

Conclusion

A PLC-controlled conveyor that automatically detects and rejects defective items was designed, programmed in 57 steps of ladder logic and validated in simulation on an FX5U. The program provides a latched run command, a 10 s start-up delay, a pause/resume function, a defect memory bit and a 2 s timed ejection pulse. The objectives set in Chapter 1 were met insimulation. The next step is hardware commissioning, followed by the extensions in the future scope.

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